

reliable & efficient neuro-symbolic AI

antonio vergari (he/him)

 @nolovedeep learning

3rd Jun 2026 - Reliable and Efficient Neuro-Symbolic AI - University of Trento

#probabilistic

#machine

#learning

3.18A

#neuro-

#symbolic

#AI

PCR students are already fully qualified professionals who perform valuable research work, albeit at a junior level. As such, they deserve to be treated just like any other working member of society and their work should be valued both within and outside of the institution. Yes, we are still learning, but we are also literally exploring new knowledge that has never been learned before – and – we're outputting it to the world with the school's and the university's name attached to it. That is a valuable contribution to both the school and the wider society. It's not a "privilege" to be paid for this work – it should be the absolute minimum expected. We deserve respect, representation and remuneration just like everyone else. We deserve access to basic social security provisions, we expect the school to value our contributions and lobby for our interests with the university and the government for the things that cannot be fixed internally.



QR code linking to the document content.

april

`april-tools.github.io`

april

about
probabilities
integrals &
logic

april

*april is
probably a
recursive
identifier of a
lab*

thanks to...



Diego Calanzone



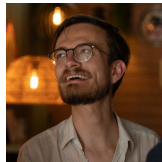
Stefano Teso



Kareem Ahmed



Lorenzo Loconte



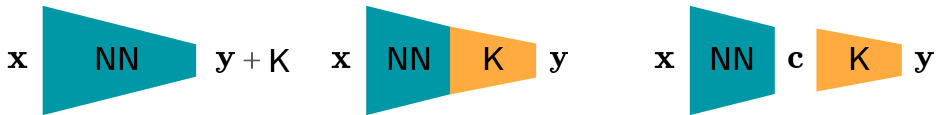
Emile van Krieken

and moar...

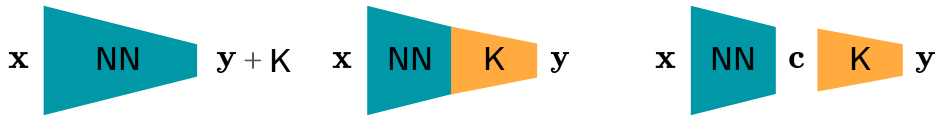
goal

of the week

***“How can neural nets
reason and learn with
symbolic constraints
reliably and efficiently?”***



***different flavours of probabilistic NeSy AI
with a single formalization***

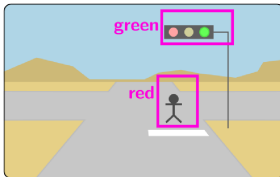


*an interplay of **logic** and **probabilistic** reasoning*

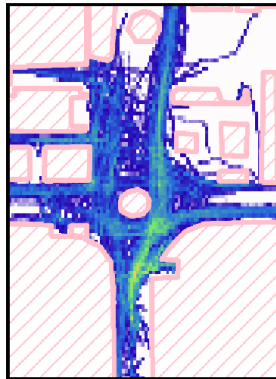
User: given the following context, generate infilling text for [BLANK] using key phrases "alien mothership", "far from over"; generated text must contain 25 - 30 words.

"First they've defeated a small squad [BLANK] are few humans left, and despite their magical power, their numbers are getting fewer."

steer text generation



hybrid driving

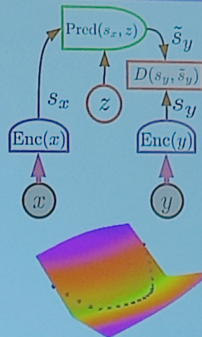


trajectory prediction

why symbolic/logic constraints?

Recommendations:

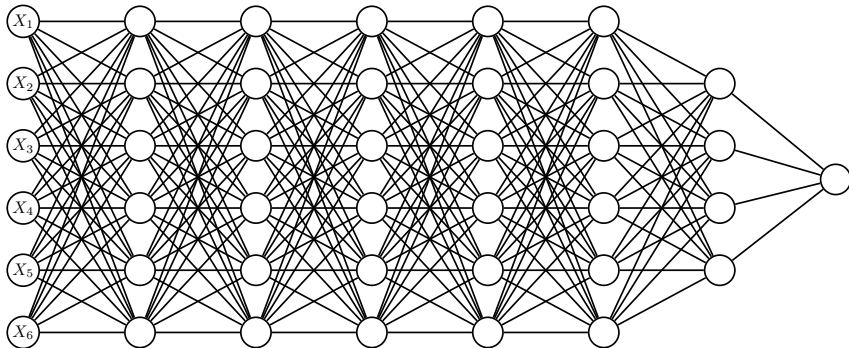
- ▶ Abandon generative models
 - ▶ in favor joint-embedding architectures
- ▶ Abandon probabilistic model
 - ▶ in favor of energy-based models
- ▶ Abandon contrastive methods
 - ▶ in favor of regularized methods
- ▶ Abandon Reinforcement Learning
 - ▶ In favor of model-predictive control
- ▶ Use RL only when planning doesn't yield the predicted outcome, to adjust the world model or the critic.



why shall we care about probabilities?

why *deep generative models*

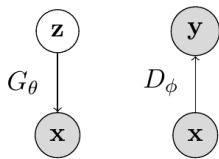
Computational graphs that encode expressive distributions $p_{\mathbf{m}}(\mathbf{X})$
 $\Rightarrow$ by stacking layers of “latent” units



why **deep generative models**

Computational graphs that encode expressive distributions $p_{\mathbf{m}}(\mathbf{X})$

$\Rightarrow$ by stacking layers of “latent” units

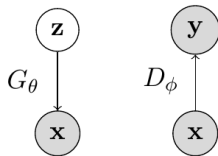


GANs

[Goodfellow et al. 2014]

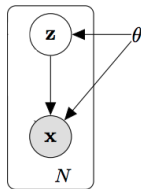
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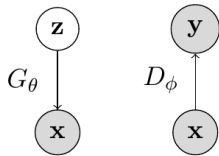


VAEs/Diffusion

[Kingma and Welling 2014]

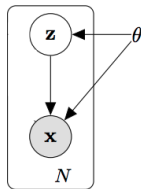
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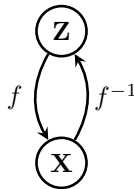
GANs

[Goodfellow et al. 2014]



VAEs/Diffusion

[Kingma and Welling 2014]



Cont/Disc Flows

[Papamakarios et al. 2019]

why **deep generative models** ...

[PROS] they encode *expressive* distributions with *millions of parameters*
+
they *scale learning* to high-dimensional, large datasets via *GPU*

...and why they lack **complex reasoning**

[PROS] they encode *expressive* distributions with *millions of parameters*

+

they *scale learning* to high-dimensional, large datasets

[CONS] they are *limited to simple reasoning* routine (lack flexibility)

+

their computations come with *little or no guarantees* (unreliable)

Deep generative models

+

flexible and reliable logic + probabilistic reasoning?

What is advanced reasoning?

or the inherent trade-off of tractability vs. expressiveness

wait!

the basics first

The “**product**” rule:

$$p(\mathbf{X}, \mathbf{Y}, \mathbf{Z}) = p(\mathbf{X})p(\mathbf{Y}, \mathbf{Z} \mid \mathbf{X}) = p(\mathbf{X})p(\mathbf{Y} \mid \mathbf{X})p(\mathbf{Z} \mid \mathbf{X}, \mathbf{Y})$$

wait!

the basics first

The “**product**” rule:

$$p(\mathbf{X}, \mathbf{Y}, \mathbf{Z}) = p(\mathbf{X})p(\mathbf{Y}, \mathbf{Z} \mid \mathbf{X}) = p(\mathbf{X})p(\mathbf{Y} \mid \mathbf{X})p(\mathbf{Z} \mid \mathbf{X}, \mathbf{Y})$$

The “**sum**” rule:

$$p(\mathbf{X}, \mathbf{Y}) = ???$$

wait!

the basics first

The “**product**” rule:

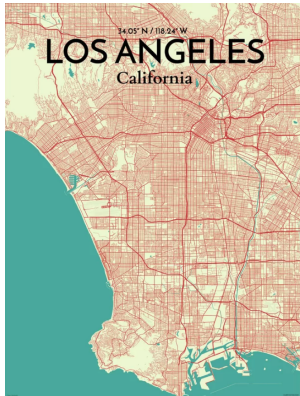
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The “**sum**” rule:

$$p(\mathbf{X}, \mathbf{Y}) = \int_{\text{dom}(\mathbf{z})} p(\mathbf{X}, \mathbf{Y}, \mathbf{z}) d\mathbf{Z}$$

Why probabilistic inference?

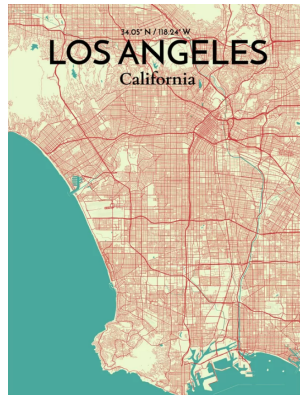
q₁: *What is the probability that today is a Monday and there is a traffic jam on Westwood Blvd.?*



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Why probabilistic inference?

- q₁**: *What is the probability that today is a Monday and there is a traffic jam on Westwood Blvd.?*
- q₂**: *Which day is most likely to have a traffic jam on my route to campus?*

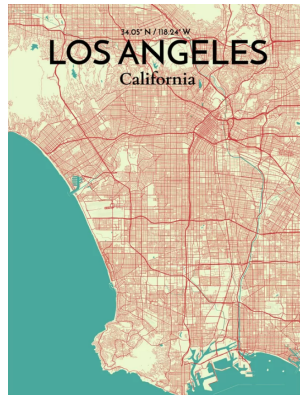


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Why probabilistic inference?

- q₁**: *What is the probability that today is a Monday and there is a traffic jam on Westwood Blvd.?*
- q₂**: *Which day is most likely to have a traffic jam on my route to campus?*

How to answer several of these **probabilistic queries**?

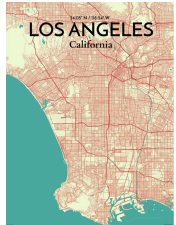


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"What is the most likely street to have a traffic jam at 12.00?"



q₁?



answering queries...

"What is the most likely street to have a traffic jam at 12.00?"

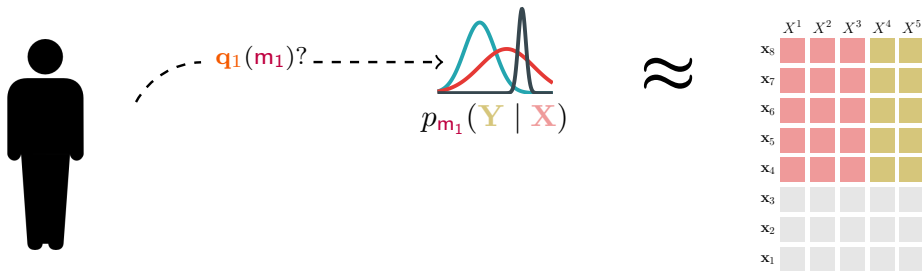


q₁?

	X^1	X^2	X^3	X^4	X^5
x_8					
x_7					
x_6					
x_5					
x_4					
x_3					
x_2					
x_1					

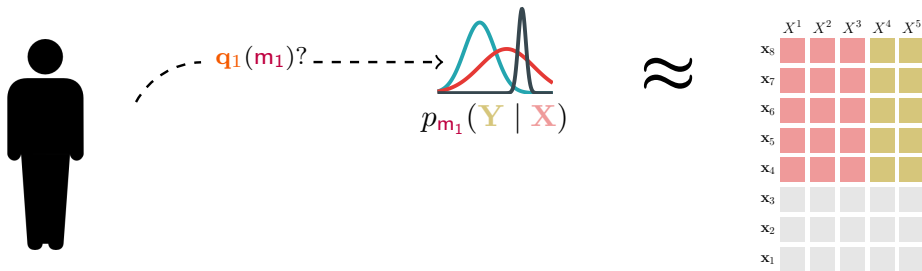
answering queries...

*"What is the most likely **street** to have a traffic jam at **12.00**?"*



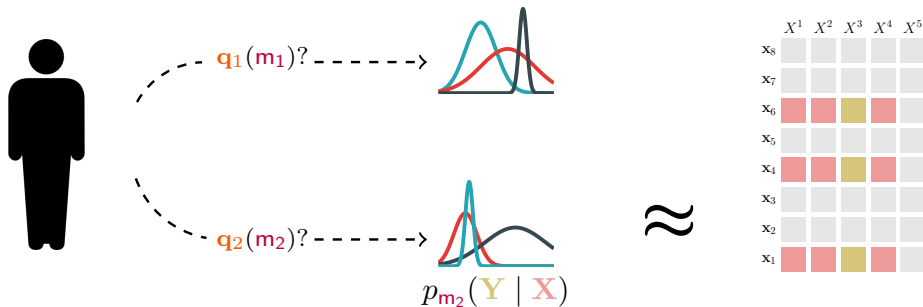
...by fitting predictive models!

*"What is the most likely **street** to have a traffic jam at **12.00**?"*



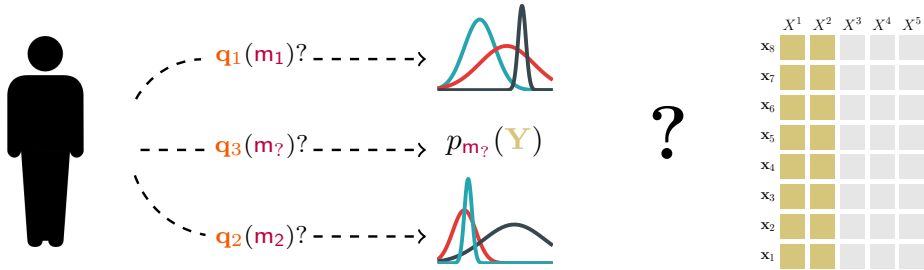
~~...by fitting predictive models!~~

"What is the most likely **time** to see a traffic jam at **Sunset Blvd.**?"

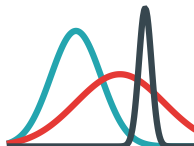
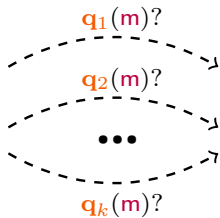


~~...by fitting predictive models!~~

"What is the probability of a traffic jam on **Westwood Blvd.** on **Monday**?"



~~...by fitting predictive models!~~

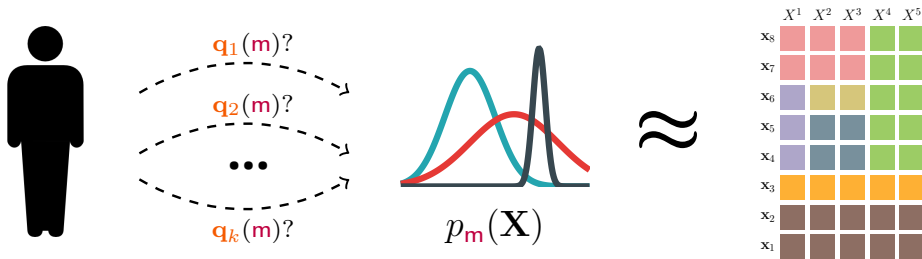


$p_m(\mathbf{X})$

$\approx$

	X^1	X^2	X^3	X^4	X^5
x_8					
x_7					
x_6					
x_5					
x_4					
x_3					
x_2					
x_1					

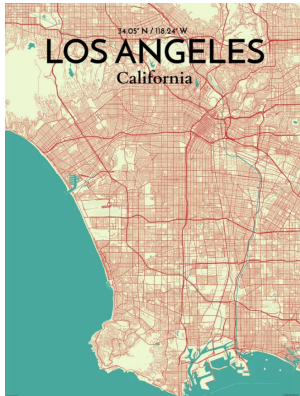
...by fitting generative models!



...e.g. exploratory data analysis

Why probabilistic inference?

q₁: *What is the probability that today is a Monday and there is a traffic jam on Westwood Blvd.?*



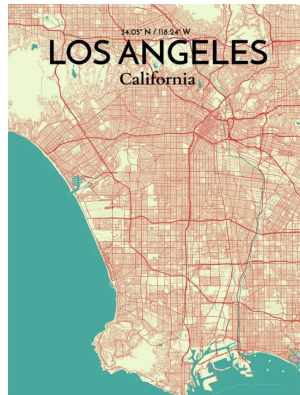
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Why probabilistic inference?

q_1 : *What is the probability that today is a Monday and there is a traffic jam on Westwood Blvd.?*

$\mathbf{X} = \{\text{Day, Time, Jam}_{\text{Str1}}, \text{Jam}_{\text{Str2}}, \dots, \text{Jam}_{\text{StrN}}\}$

$q_1(\mathbf{m}) = p_{\mathbf{m}}(\text{Day} = \text{Mon}, \text{Jam}_{\text{Wwood}} = 1)$



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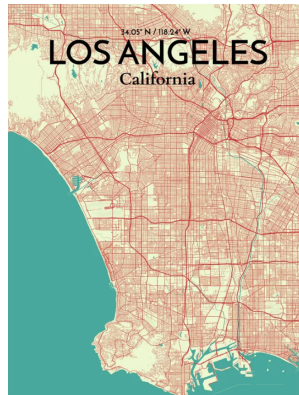
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$q_1(\mathbf{m}) = p_{\mathbf{m}}(\text{Day} = \text{Mon}, \text{Jam}_{\text{Wwood}} = 1)$

$\Rightarrow$ **marginals** (the sum rule)



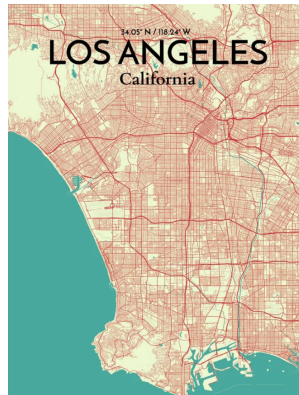
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Why probabilistic inference?

q₂: Which day is most likely to have a traffic jam on my route to campus?

$\mathbf{X} = \{\text{Day}, \text{Time}, \text{Jam}_{\text{Str1}}, \text{Jam}_{\text{Str2}}, \dots, \text{Jam}_{\text{StrN}}\}$

q₂(**m**) = $\text{argmax}_d p_{\mathbf{m}}(\text{Day} = d \wedge \bigvee_{i \in \text{route}} \text{Jam}_{\text{Str}i})$



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Why probabilistic inference?

q_2 : Which day is most likely to have a traffic jam on my route to campus?

$\mathbf{X} = \{\text{Day}, \text{Time}, \text{Jam}_{\text{Str1}}, \text{Jam}_{\text{Str2}}, \dots, \text{Jam}_{\text{StrN}}\}$

$q_2(\mathbf{m}) = \operatorname{argmax}_d p_{\mathbf{m}}(\text{Day} = d \wedge \bigvee_{i \in \text{route}} \text{Jam}_{\text{Str}i})$

$\Rightarrow$ *marginals + MAP + logical events*



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Tractable Probabilistic Inference

A class of queries $\mathcal{Q}$ is tractable on a family of probabilistic models $\mathcal{M}$
iff for any query $q \in \mathcal{Q}$ and model $m \in \mathcal{M}$
exactly computing $q(m)$ runs in time $O(\text{poly}(|m|))$.

$\Rightarrow$ **model-centric** definition...

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$\Rightarrow$ often poly will in fact be **linear**!

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$\Rightarrow$ Note: if $|m| \in O(\text{poly}(|X|))$, then query time is $O(\text{poly}(|X|))$.

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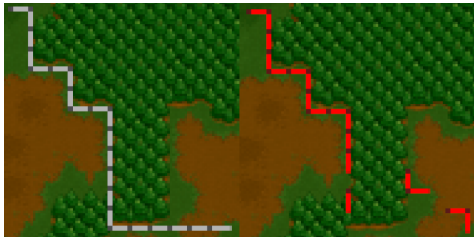
$\Rightarrow$ often poly will in fact be **linear**!

$\Rightarrow$ Note: if $|m| \in O(\text{poly}(|X|))$, then query time is $O(\text{poly}(|X|))$.

$\Rightarrow$ Why **exactness**? Highest guarantee possible!

why tractable models?

exactness can be crucial in safety-driven applications



guarantee constraint satisfaction

[Ahmed et al. 2022]



estimation error is bounded (0)

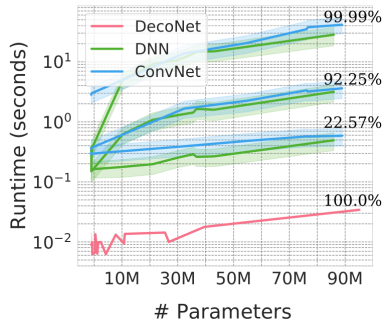
[Choi 2022]

why tractable models?

they can be much faster than intractable ones!

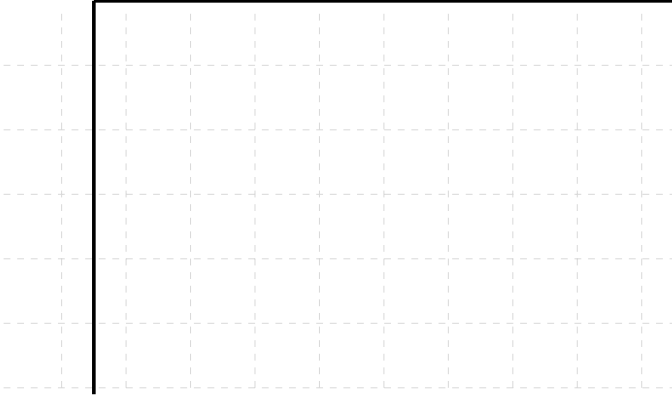
Method	MNIST (10,000 test images)		
	Theoretical bpd	Comp. bpd	En- & decoding time
PC (small)	1.26	1.30	53
PC (large)	1.20	1.24	168
IDF	1.90	1.96	880
BitSwap	1.27	1.31	904

[Liu, Mandt, and Broeck 2022]



[Subramani et al. 2021]

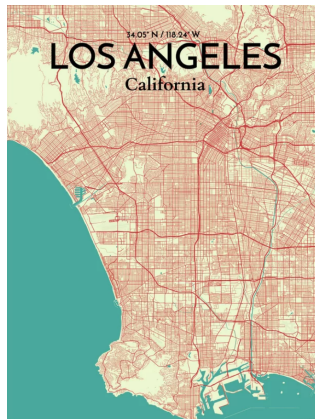
$\mathcal{M}$ $\mathcal{Q}:$



tractable bands

Complete evidence (EVI)

q₃: *What is the probability that today is a Monday at 12.00 and there is a traffic jam only on Westwood Blvd.?*



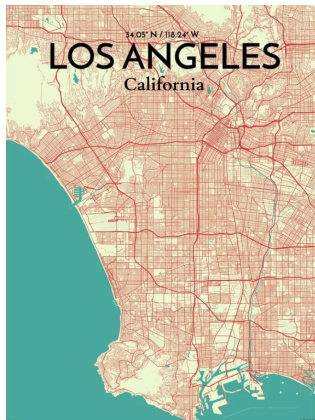
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Complete evidence (EVI)

q_3 : What is the probability that today is a Monday at 12.00 and there is a traffic jam only on Westwood Blvd.?

$\mathbf{X} = \{\text{Day, Time, Jam}_{\text{Wwood}}, \text{Jam}_{\text{Str2}}, \dots, \text{Jam}_{\text{StrN}}\}$

$q_3(\mathbf{m}) = p_{\mathbf{m}}(\mathbf{X} = \{\text{Mon, 12.00, 1, 0}, \dots, 0\})$



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Complete evidence (EVI)

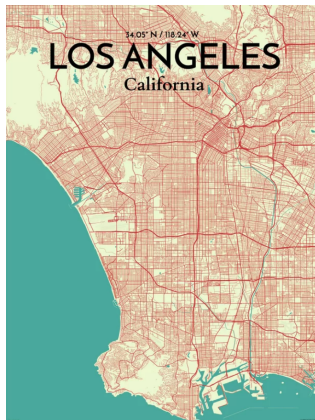
q₃: What is the probability that today is a Monday at 12.00 and there is a traffic jam only on Westwood Blvd.?

$\mathbf{X} = \{\text{Day, Time, Jam}_{\text{Wwood}}, \text{Jam}_{\text{Str2}}, \dots, \text{Jam}_{\text{StrN}}\}$

q₃(**m**) = $p_{\mathbf{m}}(\mathbf{X} = \{\text{Mon, 12.00, 1, 0}, \dots, 0\})$

...fundamental in **maximum likelihood learning**

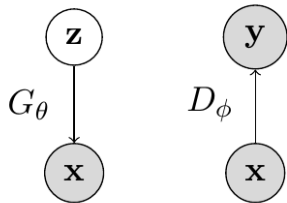
$$\theta_{\mathbf{m}}^{\text{MLE}} = \operatorname{argmax}_{\theta} \prod_{\mathbf{x} \in \mathcal{D}} p_{\mathbf{m}}(\mathbf{x}; \theta)$$



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Generative Adversarial Networks

$$\min_{\theta} \max_{\phi} \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D_{\phi}(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} [\log(1 - D_{\phi}(G_{\theta}(\mathbf{z})))]$$



*Goodfellow et al., "Generative adversarial nets",
Advances in neural information processing systems, 2014*

Generative Adversarial Networks

$$\min_{\theta} \max_{\phi} \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D_{\phi}(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} [\log(1 - D_{\phi}(G_{\theta}(\mathbf{z})))]$$

no explicit likelihood!

$\Rightarrow$ adversarial training instead of MLE

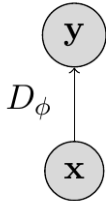
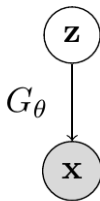
$\Rightarrow$ no tractable ELBO

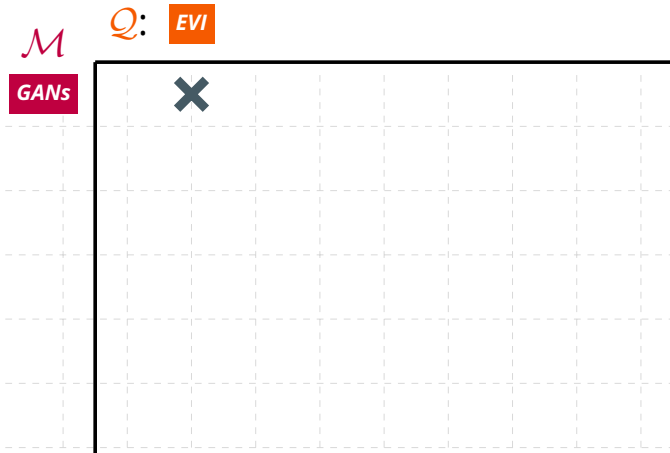
good sample quality

$\Rightarrow$ but lots of samples needed for MC

unstable training

$\Rightarrow$ mode collapse



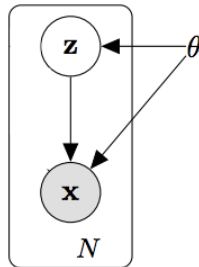


tractable bands

Variational Autoencoders

$$p_{\theta}(\mathbf{x}) = \int p_{\theta}(\mathbf{x} \mid \mathbf{z})p(\mathbf{z})d\mathbf{z}$$

an explicit likelihood model!



Rezende, Mohamed, and Wierstra, "Stochastic backprop. and approximate inference in deep generative models", , 2014

Kingma and Welling, "Auto-Encoding Variational Bayes",
Proceedings of the 2nd International Conference on Learning Representations (ICLR), 2014

Variational Autoencoders

$$\log p_{\theta}(\mathbf{x}) \geq \mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} [\log p_{\theta}(\mathbf{x} | \mathbf{z})] - \text{KL}(q_{\phi}(\mathbf{z} | \mathbf{x}) || p(\mathbf{z}))$$

an explicit likelihood model!

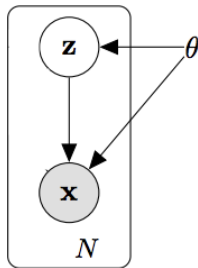
... but computing $\log p_{\theta}(\mathbf{x})$ is intractable

$\Rightarrow$ *an infinite and uncountable mixture*

$\Rightarrow$ *no tractable EVI*

we need to optimize the ELBO...

$\Rightarrow$ *which is “tricky” [Alemi et al. 2017; Dai and Wipf 2019; Ghosh et al. 2019]*



Energy Based Models

$$p_{\theta}(\mathbf{x}) = e^{-E(\mathbf{x};\theta)} / Z$$

an explicit likelihood model!

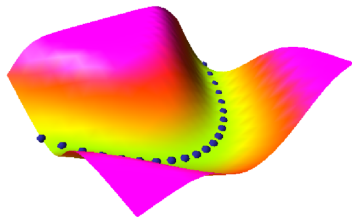
... but computing the partition function
 $Z = \int e^{-E(\mathbf{x};\theta)} d\mathbf{X}$ is intractable

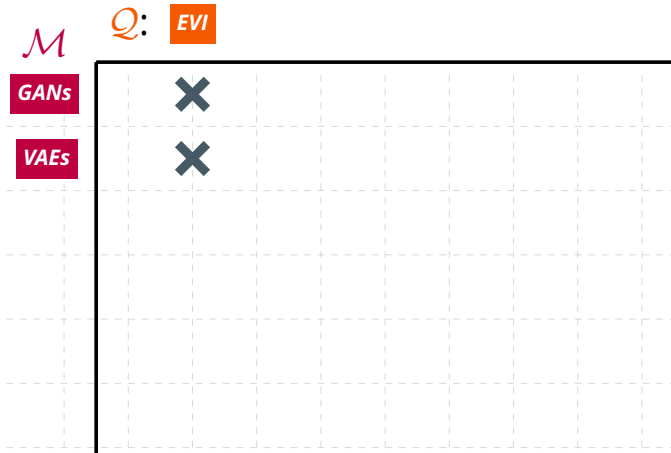
$\Rightarrow$ no tractable EVI

alternative ways to train them than MLE...

$\Rightarrow$ e.g., (denoising) score/ratio matching

[Song and Kingma 2021]





tractable bands

Normalizing flows

$$p_{\mathbf{X}}(\mathbf{x}) = p_{\mathbf{Z}}(f^{-1}(\mathbf{x})) \left| \det \left(\frac{\delta f^{-1}}{\delta \mathbf{x}} \right) \right|$$

an explicit likelihood!

...plus structured Jacobians

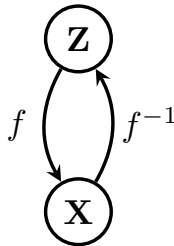
$\Rightarrow$ tractable EVI queries!

many neural variants

RealNVP [Dinh, Sohl-Dickstein, and Bengio 2016], MAF [Papamakarios, Pavlakou, and Murray 2017]

MADE [Germain et al. 2015],

PixelRNN [Oord, Kalchbrenner, and Kavukcuoglu 2016]



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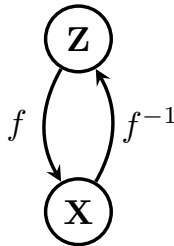
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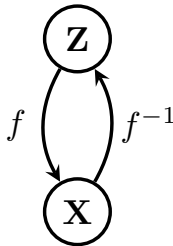
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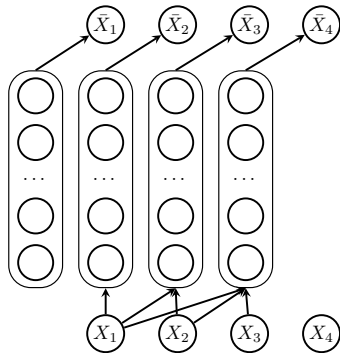


Autoregressive models

$$p_{\theta}(\mathbf{x}) = \prod_i p_{\theta}(x_i \mid \mathbf{x}_{<i})$$

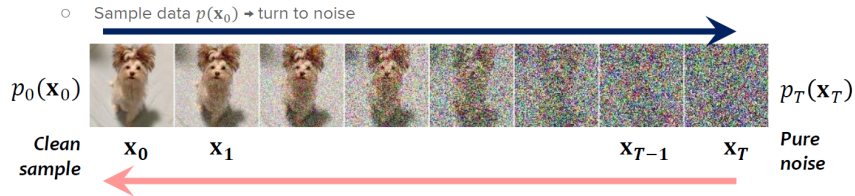
an explicit likelihood model via the product rule
a special case of normalizing flows!

$\Rightarrow$ *the $\mathbf{Z}$ s are the noise used to sample*
[Papamakarios et al. 2021]



Diffusion models

and continuous (time) normalizing flows



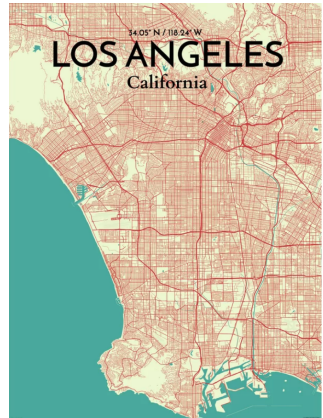
create a continuous-time flow/diffusion process as a P/SDE

$\Rightarrow$ no close form for the likelihood in general

$\Rightarrow$ no tractable EVI

Marginal queries (MAR)

q₁: What is the probability that today is a Monday ~~at~~
~~12:00~~ and there is a traffic jam ~~only~~ on Westwood
Blvd.?

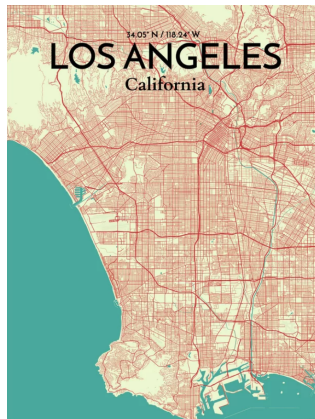


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Marginal queries (MAR)

q_1 : What is the probability that today is a Monday ~~at~~
~~12:00~~ and there is a traffic jam ~~only~~ on Westwood
Blvd.?

$$q_1(\mathbf{m}) = p_{\mathbf{m}}(\text{Day} = \text{Mon}, \text{Jam}_{\text{Wwood}} = 1)$$



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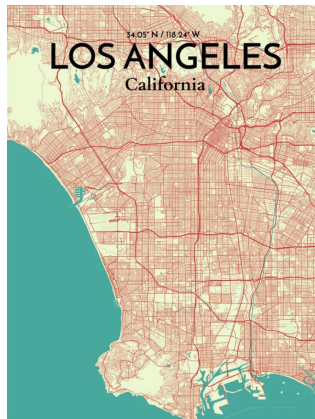
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Blvd.?

$$q_1(\mathbf{m}) = p_{\mathbf{m}}(\text{Day} = \text{Mon}, \text{Jam}_{\text{Wwood}} = 1)$$

$$\text{General: } p_{\mathbf{m}}(\mathbf{e}) = \int p_{\mathbf{m}}(\mathbf{e}, \mathbf{H}) d\mathbf{H}$$

$$\text{where } \mathbf{E} \subset \mathbf{X}, \quad \mathbf{H} = \mathbf{X} \setminus \mathbf{E}$$



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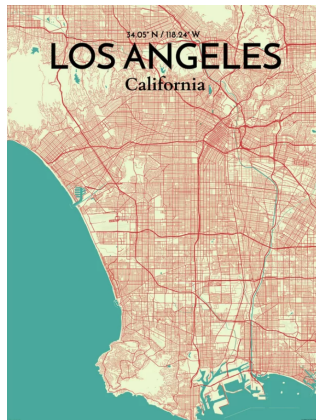
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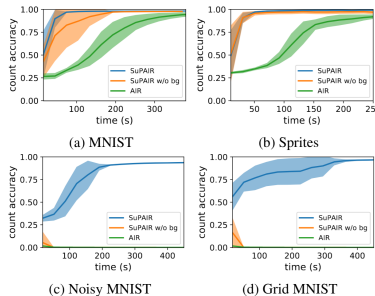
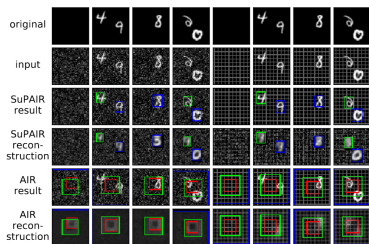
tractable MAR $\Rightarrow$ tractable **conditional queries**
(CON):

$$p_{\mathbf{m}}(\mathbf{q} \mid \mathbf{e}) = \frac{p_{\mathbf{m}}(\mathbf{q}, \mathbf{e})}{p_{\mathbf{m}}(\mathbf{e})}$$



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Tractable MAR : scene understanding



Fast and exact marginalization over unseen or “do not care” parts in the scene

Stelzner, Peharz, and Kersting, “Faster Attend-Infer-Repeat with Tractable Probabilistic Models”, Proceedings of the 36th International Conference on Machine Learning, 2019

Kossen et al., “Structured Object-Aware Physics Prediction for Video Modeling and Planning”, , 2019

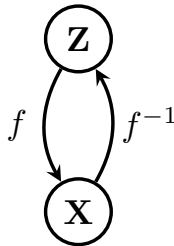
Normalizing flows

$$p_{\mathbf{X}}(\mathbf{x}) = p_{\mathbf{Z}}(f^{-1}(\mathbf{x})) \left| \det \left(\frac{\delta f^{-1}}{\delta \mathbf{x}} \right) \right|$$

an explicit likelihood!

...plus structured Jacobians

$\Rightarrow$ tractable EVI queries!



Normalizing flows

$$p_{\mathbf{X}}(\mathbf{x}) = p_{\mathbf{Z}}(f^{-1}(\mathbf{x})) \left| \det \left(\frac{\delta f^{-1}}{\delta \mathbf{x}} \right) \right|$$

an explicit likelihood!

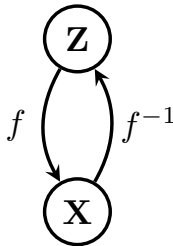
...plus structured Jacobians

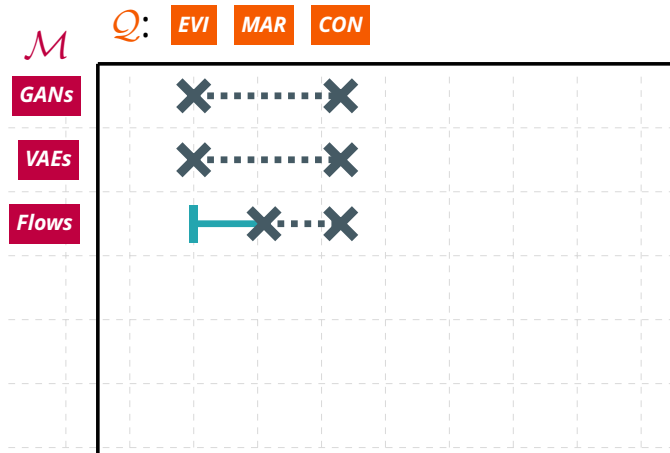
$\Rightarrow$ tractable EVI queries!

MAR is generally intractable:

we cannot easily integrate over f

$\Rightarrow$ unless f is "simple", e.g. identity





tractable bands

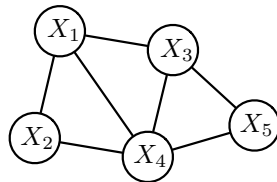
Probabilistic Graphical Models (PGMs)

Declarative semantics: a clean separation of modeling assumptions from inference

Nodes: random variables

Edges: dependencies

+



Inference: conditioning *[Darwiche 2001; Sang, Beame, and Kautz 2005]*

elimination *[Zhang and Poole 1994; Dechter 1998]*

message passing *[Yedidia, Freeman, and Weiss*



MOGWAI

PGMs WILL NEVER DIE,
BUT YOU WILL.

Complexity of MAR on PGMs

Exact complexity: Computing MAR and CON is *#P-hard*

$\Rightarrow$ [Cooper 1990; Roth 1996]

Approximation complexity: Computing MAR and CON approximately within a relative error of $2^{n^{1-\epsilon}}$ for any fixed ϵ is *NP-hard*

$\Rightarrow$ [Dagum and Luby 1993; Roth 1996]

Why? Treewidth!

Treewidth:

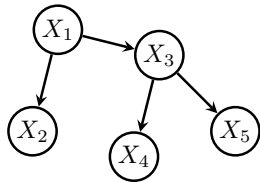
Informally, how tree-like is the graphical model $\mathbf{m}$?

Formally, the minimum width of any tree-decomposition of $\mathbf{m}$.

Fixed-parameter tractable: MAR and CON on a graphical model $\mathbf{m}$ with treewidth w take time $O(|\mathbf{X}| \cdot 2^w)$, which is linear for fixed width w

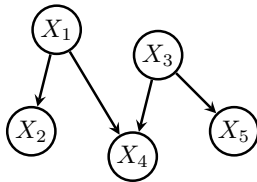
[Dechter 1998; Koller and Friedman 2009]. $\Rightarrow$ what about bounding the treewidth by design?

Low-treewidth PGMs



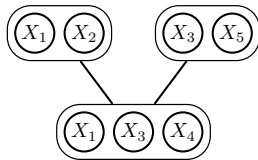
Trees

[Meilă and Jordan 2000]



Polytrees

[Dasgupta 1999]



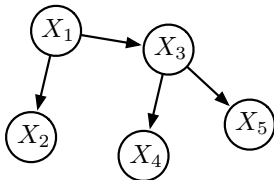
Thin Junction trees

[Bach and Jordan 2001]

If treewidth is bounded (e.g. ≈ 20), exact MAR and CON inference is possible in practice

Tree distributions

A **tree-structured BN** [Meilă and Jordan 2000] where each $X_i \in \mathbf{X}$ has *at most* one parent Pa_{X_i} .

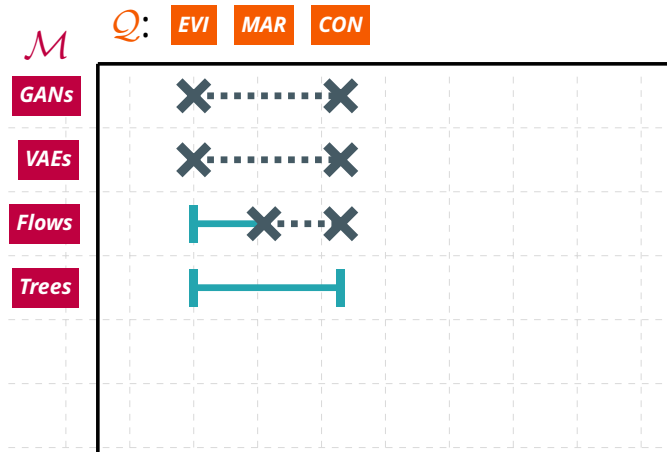


$$p(\mathbf{X}) = \prod_{i=1}^n p(x_i | \text{Pa}_{x_i})$$

Exact querying: EVI, MAR, CON tasks *linear* for trees: $O(|\mathbf{X}|)$

Exact learning from d examples takes $O(|\mathbf{X}|^2 \cdot d)$ with the classical Chow-Liu algorithm¹

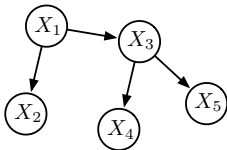
¹Chow and Liu, "Approximating discrete probability distributions with dependence trees", 1968



tractable bands

What do we lose?

Expressiveness: Ability to represent rich and complex classes of distributions



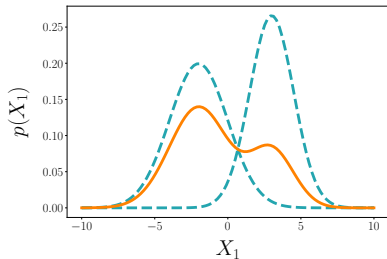
Bounded-treewidth PGMs lose the ability to represent *all possible distributions* ...

Cohen, Sharir, and Shashua, "On the expressive power of deep learning: A tensor analysis",
Conference on Learning Theory, 2016

Martens and Medabalimi, "On the Expressive Efficiency of Sum Product Networks", , 2014

Mixtures

Mixtures as a convex combination of k (simpler) probabilistic models

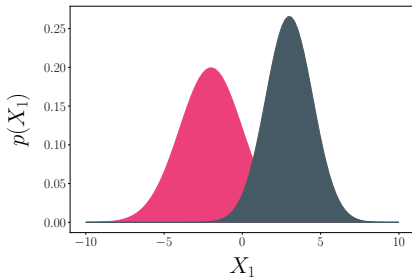


$$p(X) = w_1 \cdot p_1(X) + w_2 \cdot p_2(X)$$

EVI, MAR, CON queries scale linearly in k

Mixtures

Mixtures as a convex combination of k (simpler) probabilistic models



$$p(X) = p(Z = 1) \cdot p_1(X|Z = 1) \\ + p(Z = 2) \cdot p_2(X|Z = 2)$$

Mixtures are marginalizing a **categorical latent variable** Z with k values

$\Rightarrow$ increased expressiveness

Expressiveness and efficiency

Expressiveness: Ability to represent rich and effective classes of functions

$\Rightarrow$ *mixture of Gaussians can approximate any distribution!*

*Cohen, Sharir, and Shashua, "On the expressive power of deep learning: A tensor analysis",
Conference on Learning Theory, 2016*

Martens and Medabalimi, "On the Expressive Efficiency of Sum Product Networks", , 2014

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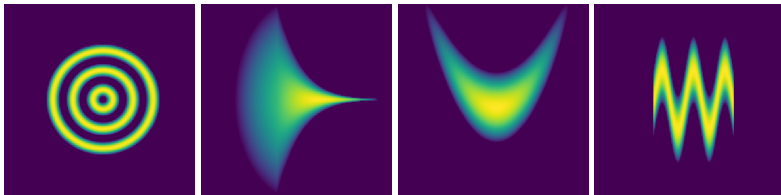
Expressive efficiency (succinctness) Ability to represent rich and effective classes of functions **compactly**

$\Rightarrow$ *but how many components does a Gaussian mixture need?*

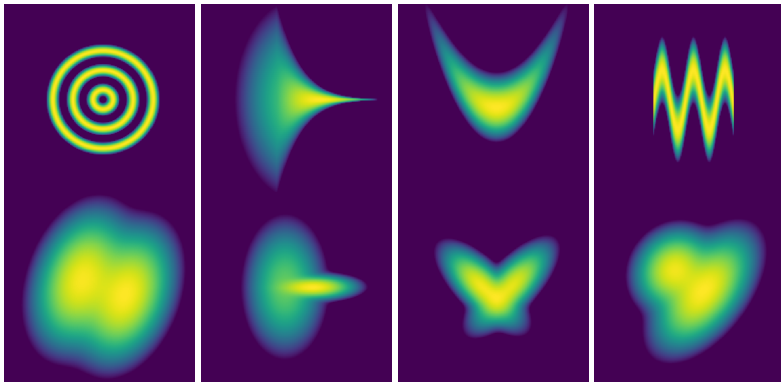
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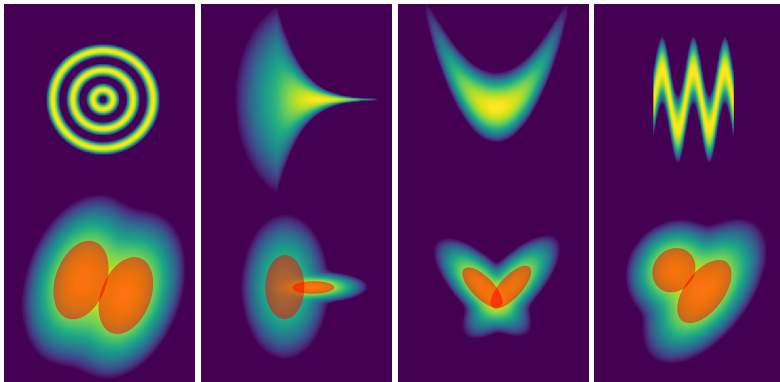
How expressive efficient are mixtures?



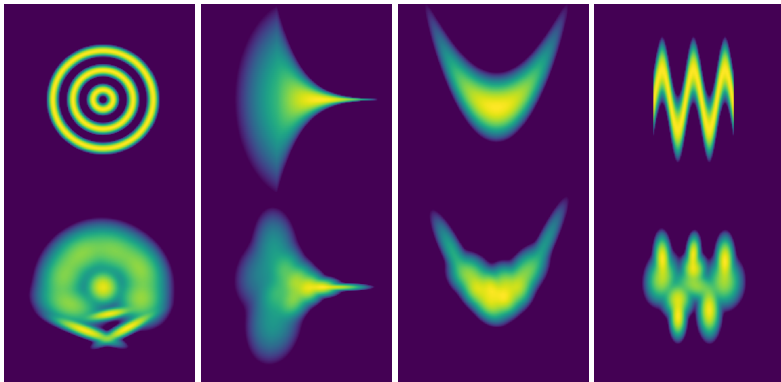
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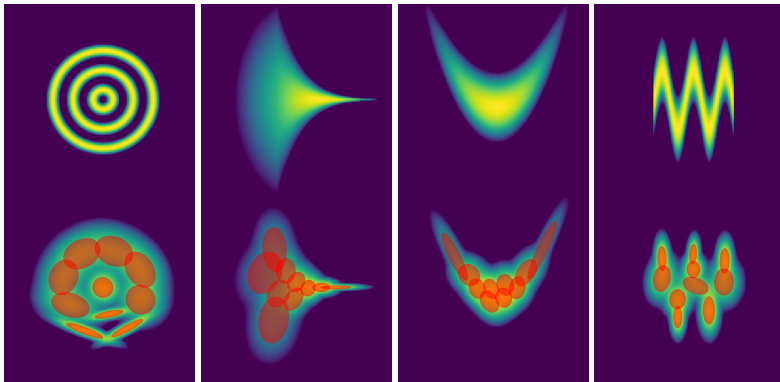
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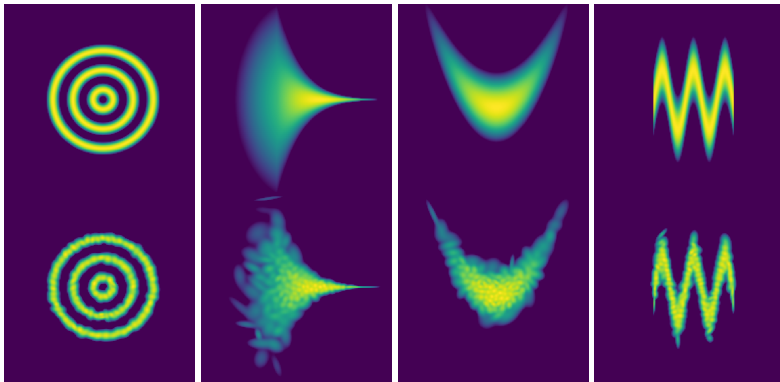
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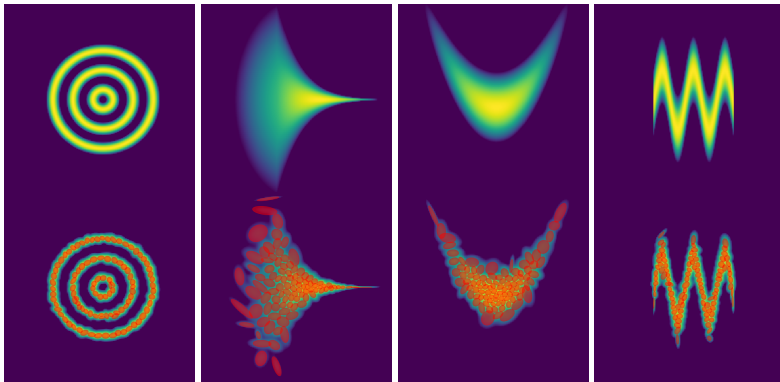
How expressive efficient are mixtures?



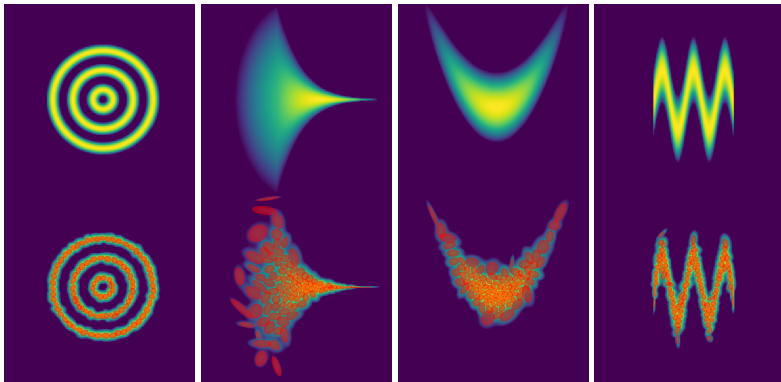
How expressive efficient are mixtures?



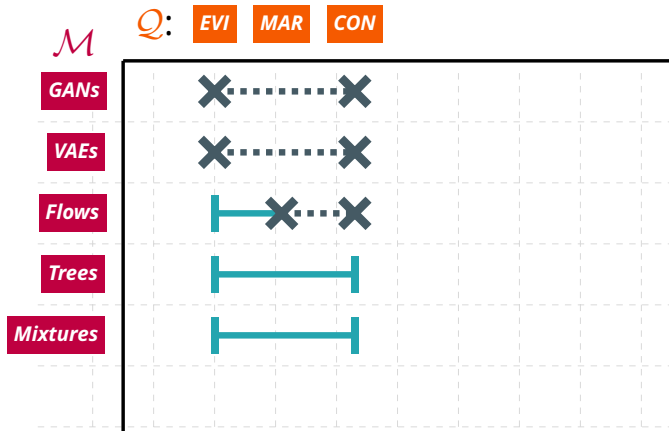
How expressive efficient are mixtures?



How expressive efficient are mixtures?



⇒ *stack mixtures like in deep generative models* **46/67**

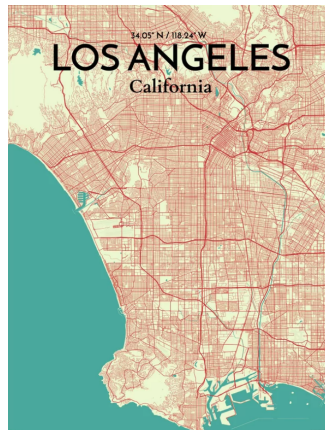


tractable bands

Maximum A Posteriori (MAP)

aka Most Probable Explanation (MPE)

q₅: Which combination of roads is most likely to be jammed on Monday at 9am?



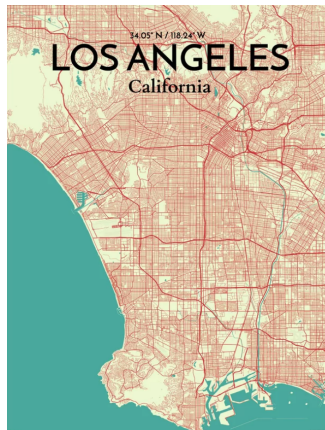
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Maximum A Posteriori (MAP)

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q₅: Which combination of roads is most likely to be jammed on Monday at 9am?

$$\mathbf{q}_5(\mathbf{m}) = \operatorname{argmax}_{\mathbf{j}} p_{\mathbf{m}}(\mathbf{j}_1, \mathbf{j}_2, \dots \mid \text{Day} = \text{M}, \text{Time} = 9)$$



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Maximum A Posteriori (MAP)

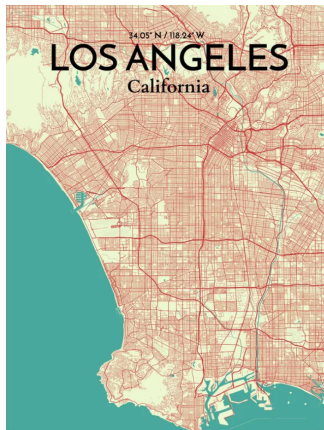
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General: $\operatorname{argmax}_{\mathbf{q}} p_{\mathbf{m}}(\mathbf{q} \mid \mathbf{e})$

where $\mathbf{Q} \cup \mathbf{E} = \mathbf{X}$



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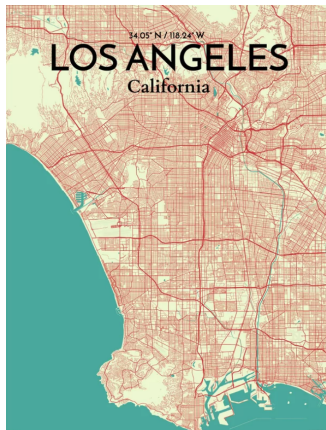
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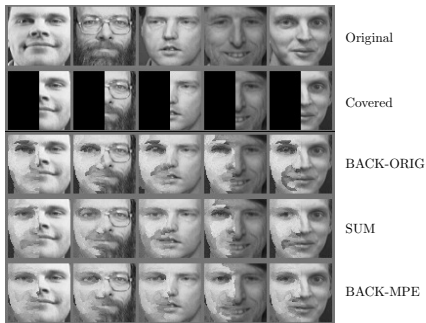
...**intractable** for latent variable models!

$$\begin{aligned}\max_{\mathbf{q}} p_{\mathbf{m}}(\mathbf{q} \mid \mathbf{e}) &= \max_{\mathbf{q}} \sum_{\mathbf{z}} p_{\mathbf{m}}(\mathbf{q}, \mathbf{z} \mid \mathbf{e}) \\ &\neq \sum_{\mathbf{z}} \max_{\mathbf{q}} p_{\mathbf{m}}(\mathbf{q}, \mathbf{z} \mid \mathbf{e})\end{aligned}$$



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MAP inference : image inpainting

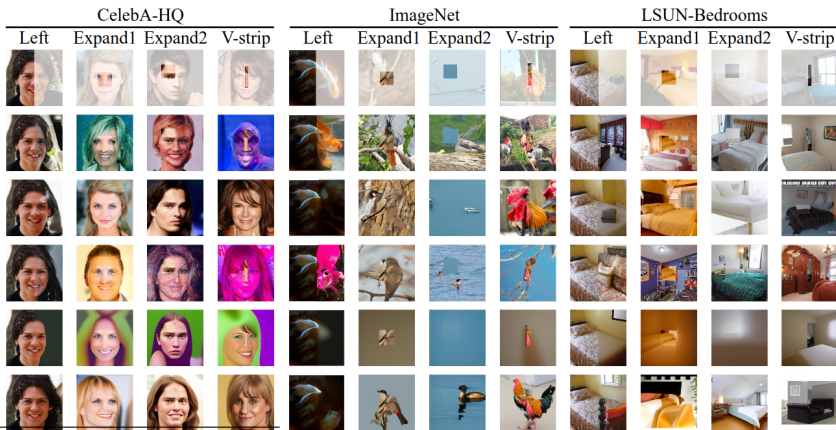


Predicting **arbitrary patches**
given a **single** model
without the need of retraining.

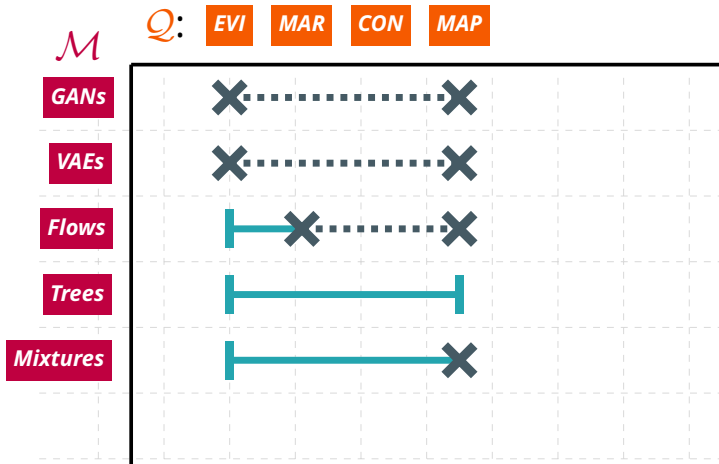
Poon and Domingos, "Sum-Product Networks: a New Deep Architecture", , 2011

Sguerra and Cozman, "Image classification using sum-product networks for autonomous flight of micro aerial vehicles", 2016 5th Brazilian Conference on Intelligent Systems (BRACIS), 2016

MAP inference : image inpainting



Liu, Niepert, and Broeck, "Image Inpainting via Tractable Steering of Diffusion Models", *ICLR*, 2024

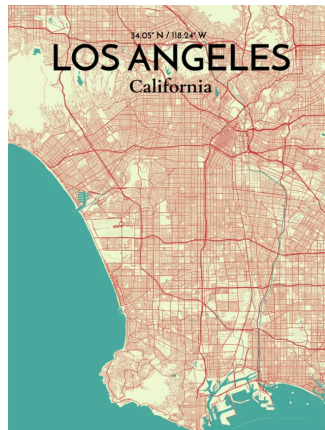


tractable bands

Marginal MAP (MMAP)

aka Bayesian Network MAP

q₆: Which combination of roads is most likely to be jammed ~~on Monday~~ at 9am?



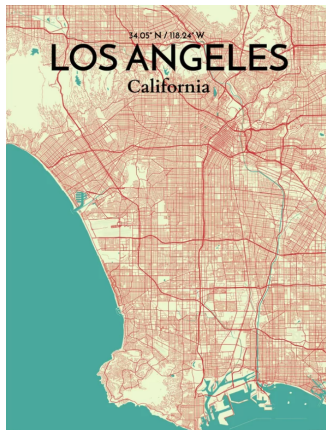
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Marginal MAP (MMAP)

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q₆: Which combination of roads is most likely to be jammed ~~on Monday~~ at 9am?

$$\mathbf{q}_6(\mathbf{m}) = \operatorname{argmax}_{\mathbf{j}} p_{\mathbf{m}}(\mathbf{j}_1, \mathbf{j}_2, \dots \mid \text{Time}=9)$$



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Marginal MAP (MMAP)

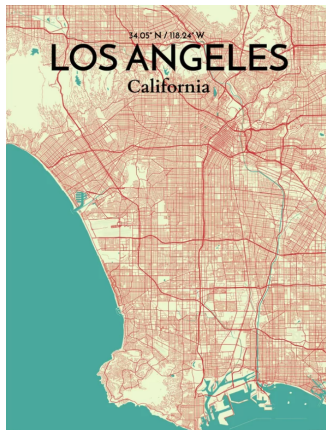
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$$\begin{aligned} \text{General: } \operatorname{argmax}_{\mathbf{q}} p_{\mathbf{m}}(\mathbf{q} \mid \mathbf{e}) \\ = \operatorname{argmax}_{\mathbf{q}} \sum_{\mathbf{h}} p_{\mathbf{m}}(\mathbf{q}, \mathbf{h} \mid \mathbf{e}) \end{aligned}$$

$$\text{where } \mathbf{Q} \cup \mathbf{H} \cup \mathbf{E} = \mathbf{X}$$



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Marginal MAP (MMAP)

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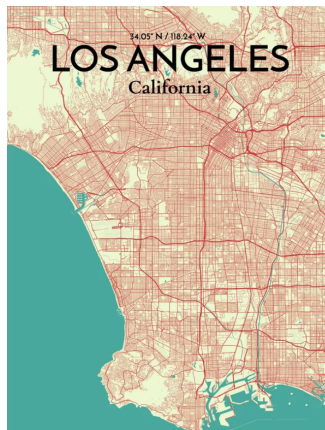
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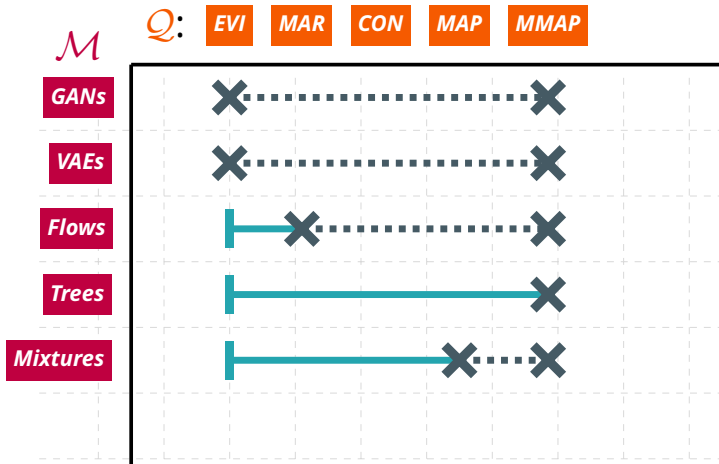
$$\mathbf{q}_6(\mathbf{m}) = \operatorname{argmax}_{\mathbf{j}} p_{\mathbf{m}}(\mathbf{j}_1, \mathbf{j}_2, \dots \mid \text{Time}=9)$$

⇒ NP^{PP}-complete [Park and Darwiche 2006]

⇒ NP-hard for trees [de Campos 2011]

⇒ NP-hard even for Naive Bayes [ibid.]

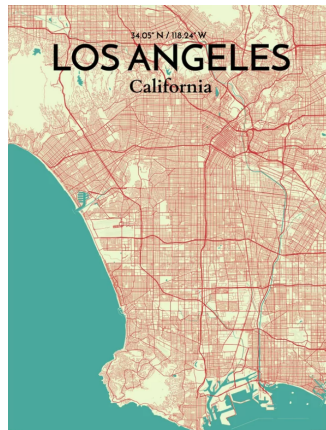




tractable bands

Advanced queries

q₂: Which day is most likely to have a traffic jam on my route to campus?



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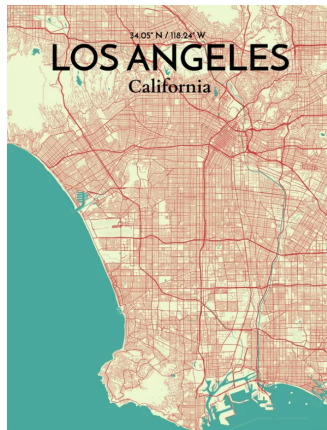
*Bekker et al., "Tractable Learning for Complex Probability Queries",
Advances in Neural Information Processing Systems 28 (NIPS), 2015*

Advanced queries

q₂: Which day is most likely to have a traffic jam on my route to campus?

$$q_2(\mathbf{m}) = \operatorname{argmax}_d p_{\mathbf{m}}(\text{Day} = d \wedge \bigvee_{i \in \text{route}} \text{Jam}_{\text{Str } i})$$

⇒ **marginals + MAP + logical events**

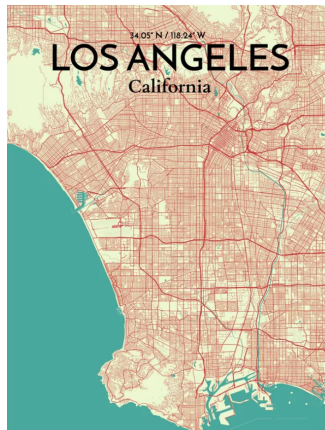


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*Bekker et al., "Tractable Learning for Complex Probability Queries",
Advances in Neural Information Processing Systems 28 (NIPS), 2015*

Advanced queries

- q₂**: Which day is most likely to have a traffic jam on my route to campus?
- q₇**: What is the probability of seeing more traffic jams in Westwood than Hollywood?



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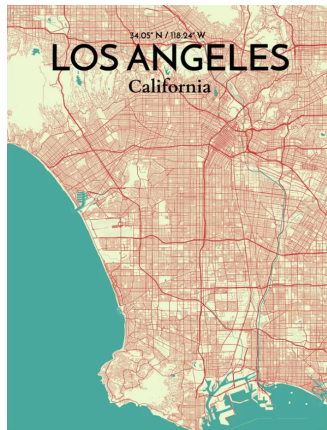
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Advances in Neural Information Processing Systems 28 (NIPS), 2015*

Advanced queries

q₂: Which day is most likely to have a traffic jam on my route to campus?

q₇: What is the probability of seeing more traffic jams in Westwood than Hollywood?

⇒ **counts + group comparison**



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*Bekker et al., "Tractable Learning for Complex Probability Queries",
Advances in Neural Information Processing Systems 28 (NIPS), 2015*

Advanced queries

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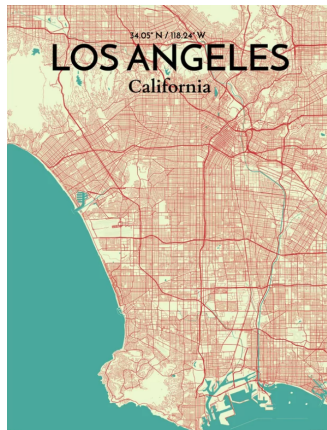
and more:

expected classification agreement

[Oztok, Choi, and Darwiche 2016; Choi, Darwiche, and Broeck 2017; Choi and Broeck 2018]

expected predictions *[Khosravi et al. 2019]*

Bekker et al., "Tractable Learning for Complex Probability Queries", Advances in Neural Information Processing Systems 28 (NIPS), 2015



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more complex reasoning



q₁

"What is the **expected prediction** for a patient with unavailable records?"



q₂

"How **fair** is the prediction is a certain protected attribute changes?"



q₃

"Can we certify no **adversarial examples** exist?"

*...asking **queries** to a ML model*

more complex reasoning



q₁

$\mathbb{E}_{\mathbf{x}_m \sim p(\mathbf{x}_m | \mathbf{x}_o)} [f(\mathbf{x}_o, \mathbf{x}_m)]$
(*expected prediction*)

q₂

$\mathbb{E}_{\mathbf{x}_c \sim p(\mathbf{x}_c | X_s=0)} [f_0(\mathbf{x}_c)] -$
 $\mathbb{E}_{\mathbf{x}_c \sim p(\mathbf{x}_c | X_s=1)} [f_1(\mathbf{x}_c)]$
(*fairness*)

q₃

$\mathbb{E}_{\mathbf{e} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)} [f(\mathbf{x} + \mathbf{e})]$
(*adversarial robust.*)

...into math expressions

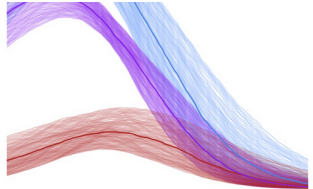
more complex reasoning



neuro-symbolic AI



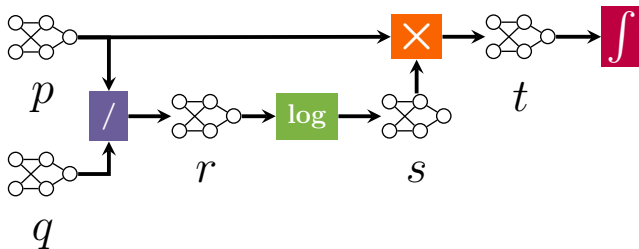
probabilistic programming



computing uncertainties
(Bayesian inference)

...and more application scenarios

$$\int p(\mathbf{x}) \times \log \left(p(\mathbf{x}) / q(\mathbf{x}) \right) d\mathbf{X}$$



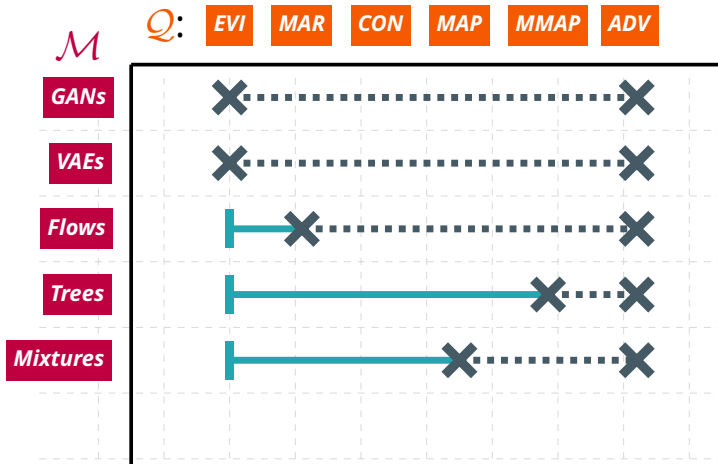
build a LEGO-like query calculus!

	Query	Tract. Conditions	Hardness
CROSS ENTROPY	$-\int p(\mathbf{x}) \log q(\mathbf{x}) d\mathbf{X}$	Cmp, q Det	#P-hard w/o Det
SHANNON ENTROPY	$-\sum p(\mathbf{x}) \log p(\mathbf{x})$	Sm, Dec, Det	coNP-hard w/o Det
RÉNYI ENTROPY	$(1 - \alpha)^{-1} \log \int p^\alpha(\mathbf{x}) d\mathbf{X}, \alpha \in \mathbb{N}$	SD	#P-hard w/o SD
MUTUAL INFORMATION	$(1 - \alpha)^{-1} \log \int p^\alpha(\mathbf{x}) d\mathbf{X}, \alpha \in \mathbb{R}_+$	Sm, Dec, Det	#P-hard w/o Det
KULLBACK-LEIBLER DIV.	$\int p(\mathbf{x}, \mathbf{y}) \log(p(\mathbf{x}, \mathbf{y}) / (p(\mathbf{x})p(\mathbf{y})))$	Sm, SD, Det*	coNP-hard w/o SD
RÉNYI'S ALPHA DIV.	$\int p(\mathbf{x}) \log(p(\mathbf{x}) / q(\mathbf{x})) d\mathbf{X}$	Cmp, Det	#P-hard w/o Det
ITAKURA-SAITO DIV.	$(1 - \alpha)^{-1} \log \int p^\alpha(\mathbf{x}) q^{1-\alpha}(\mathbf{x}) d\mathbf{X}, \alpha \in \mathbb{N}$	Cmp, q Det	#P-hard w/o Det
CAUCHY-SCHWARZ DIV.	$(1 - \alpha)^{-1} \log \int p^\alpha(\mathbf{x}) q^{1-\alpha}(\mathbf{x}) d\mathbf{X}, \alpha \in \mathbb{R}$	Cmp, Det	#P-hard w/o Det
SQUARED LOSS	$\int [p(\mathbf{x}) / q(\mathbf{x}) - \log(p(\mathbf{x}) / q(\mathbf{x})) - 1] d\mathbf{X}$	Cmp, Det	#P-hard w/o Det
	$-\log \frac{\int p(\mathbf{x}) q(\mathbf{x}) d\mathbf{X}}{\sqrt{\int p^2(\mathbf{x}) d\mathbf{X} \int q^2(\mathbf{x}) d\mathbf{X}}}$	Cmp	#P-hard w/o Cmp
	$\int (p(\mathbf{x}) - q(\mathbf{x}))^2 d\mathbf{X}$	Cmp	#P-hard w/o Cmp

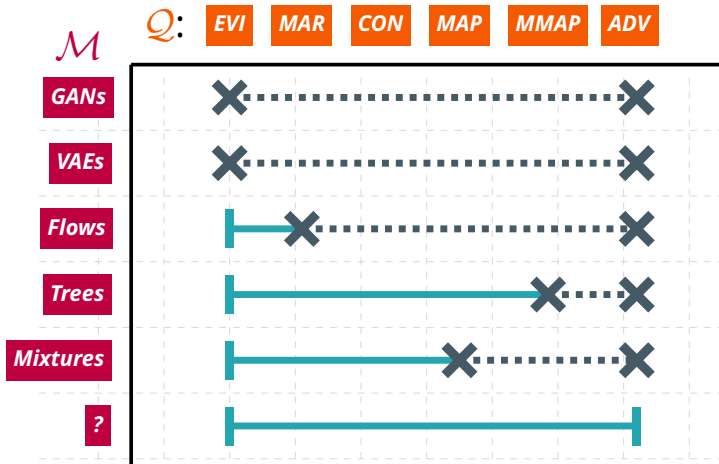
compositionally derive the tractability of many more queries

	Query	Tract. Conditions	Hardness
CROSS ENTROPY	$-\int p(\mathbf{x}) \log q(\mathbf{x}) d\mathbf{X}$	Cmp, q Det	#P-hard w/o Det
SHANNON ENTROPY	$-\sum p(\mathbf{x}) \log p(\mathbf{x})$	Sm, Dec, Det	coNP-hard w/o Det
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KULLBACK-LEIBLER DIV.	$\int p(\mathbf{x}, \mathbf{y}) \log(p(\mathbf{x}, \mathbf{y}) / (p(\mathbf{x})p(\mathbf{y})))$	Sm, SD, Det*	coNP-hard w/o SD
RÉNYI'S ALPHA DIV.	$\int p(\mathbf{x}) \log(p(\mathbf{x}) / q(\mathbf{x})) d\mathbf{X}$	Cmp, Det	#P-hard w/o Det
ITAKURA-SAITO DIV.	$(1 - \alpha)^{-1} \log \int p^\alpha(\mathbf{x}) q^{1-\alpha}(\mathbf{x}) d\mathbf{X}, \alpha \in \mathbb{N}$	Cmp, q Det	#P-hard w/o Det
CAUCHY-SCHWARZ DIV.	$(1 - \alpha)^{-1} \log \int p^\alpha(\mathbf{x}) q^{1-\alpha}(\mathbf{x}) d\mathbf{X}, \alpha \in \mathbb{R}$	Cmp, Det	#P-hard w/o Det
SQUARED LOSS	$\int [p(\mathbf{x}) / q(\mathbf{x}) - \log(p(\mathbf{x}) / q(\mathbf{x})) - 1] d\mathbf{X}$	Cmp, Det	#P-hard w/o Det
	$-\log \frac{\int p(\mathbf{x}) q(\mathbf{x}) d\mathbf{X}}{\sqrt{\int p^2(\mathbf{x}) d\mathbf{X} \int q^2(\mathbf{x}) d\mathbf{X}}}$	Cmp	#P-hard w/o Cmp
	$\int (p(\mathbf{x}) - q(\mathbf{x}))^2 d\mathbf{X}$	Cmp	#P-hard w/o Cmp

*and **prove hardness** when some input properties are not satisfied*



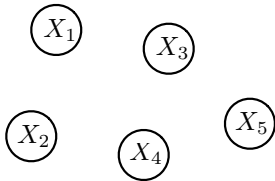
tractable bands



tractable bands

Fully factorized models

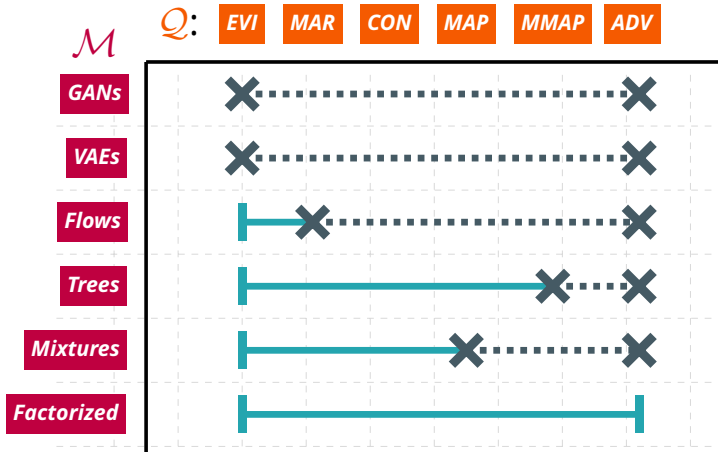
A completely disconnected graph. Example: Product of Bernoullis (PoBs)



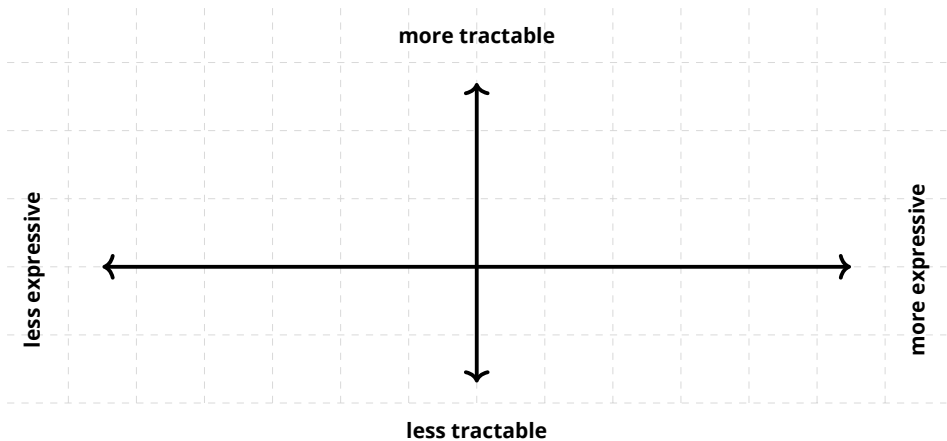
$$p(\mathbf{x}) = \prod_{i=1}^n p(x_i)$$

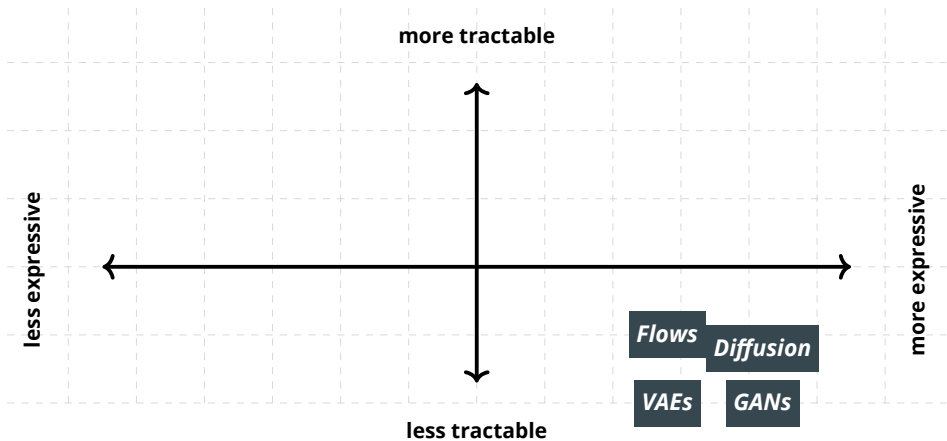
Complete evidence, marginals and MAP, MMAP inference is **linear**!

$\Rightarrow$ *but definitely not expressive...*

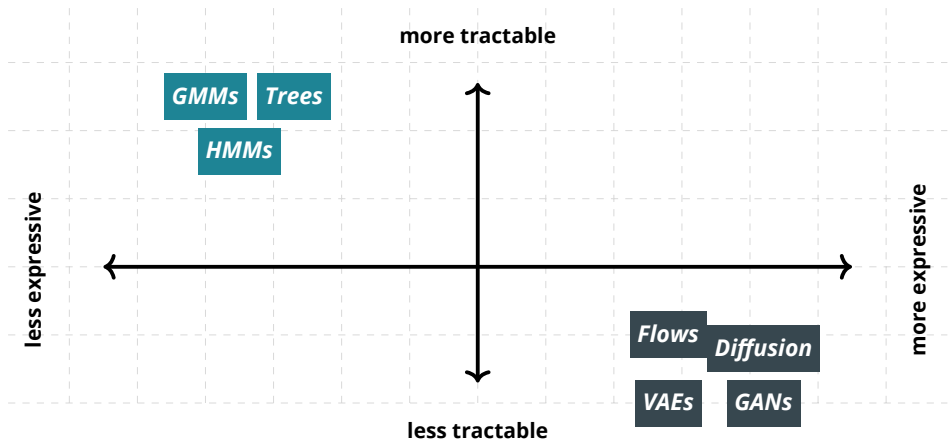


tractable bands

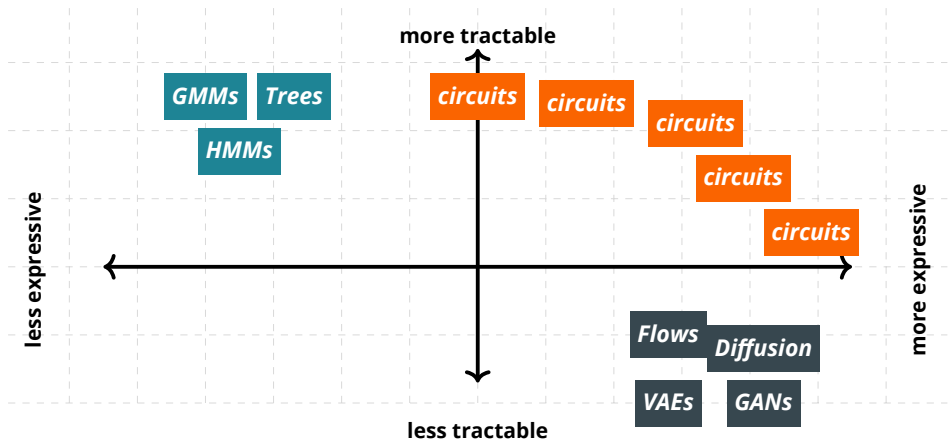




Expressive models are not much tractable...



Tractable models are not that expressive...



navigate the trade-off